

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	M. Soundarya	Reg. No.:	192524342
Section / Batch:	slot - A	Date:	31/07/26

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type		Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

34. Binary search : Even number sorted dataset .
 The sorted dataset [4 , 8 , 12 , 16 , 20 , 24 , 28 , 32]
 Binary search to find the element 20 .

Binary search - Even Number sorted Dataset.Given:

The sorted dataset [4, 8, 12, 16, 20, 24, 28, 32]
Binary search to find element 20.

Index:	0	1	2	3	4	5	6	7
Value:	4	8	12	16	20	24	28	32

Target element = 20

Binary search steps.

Initial values

* Low = 0

* High = 7

Step 1

$$\text{mid} = (0 + 7) / 2 = 3$$

Element at mid = 16

Since $20 > 16$, search right half.

New values:

* Low = 4

* High = 7

step 2

$$\text{Mid} = (4+7)/2 = 5$$

element at mid = 24

since $20 < 24$, search left half.

New values:

$$\text{Low} = 4$$

$$\text{High} = 4$$

Step 3

$$\text{Mid} = (4+4)/2 = 4$$

element at mid = 20

Element found.

Numbers of Comparisons

Comparison 1 $\rightarrow$ 16

Comparison 2 $\rightarrow$ 24

Comparison 3 $\rightarrow$ 20 (found)

Time Complexity

Best case: $O(1)$

Average case: $O(\log n)$

Worst case: $O(\log n)$

For $n=8$,

$$\log_2(8) = 3$$

The search required only 3 comparisons

Correctness of Result.

The target element 20 is found at index 4.

Output:

Element 20 found at index 4

Total comparisons = 3

Efficiency Interpretation.

For this dataset 8 elements. Only 3 comparisons were needed instead of checking each element one by one. Binary search is highly efficient for sorted datasets because it eliminates half of remaining elements.

Final Answer:

Element found: 20

Index: 4

Comparison: 3.

Code of Conduct1 M. SoundarajaC192524342

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: [Signature] M.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)	4	4	4	4	4	40
Q2 (30 Marks)	4	4	4	4	4	30
Q3 (30 Marks)	4	3	3	2	2	22
Overall Total (100)						92

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____